

ELITE MATHEMATICS

Pearson Edexcel International GCSE

Linear 4MA1

Adaptive Classified

OL-T20 · Completing the Square

Questions

Student book · questions with writing space

■ 10 pages

FAMILY MAP

Completing the Square

F01

Complete the square for a non-monic quadratic

8 Q | 24 marks

F02

Complete the square for a monic quadratic

1 Q | 3 marks

Working Unit 27

4MA1-2H Q25

UNIT 1

3 marks

25 (a) Write $28 + 24x - 6x^2$ in the form $a - b(x - c)^2$ where a , b and c are integers.

.....
(3)

Question 16

4MA1-1H Q25

MODULAR UNIT 1

4 marks

25 $f(x) = 17 - 3x^2 + 12x$

Write $f(x)$ in the form $a - b(x - c)^2$ where a , b and c are constants. $f(x) = \dots\dots\dots$

4MA1-2H Q19

3 marks

.....

Question 15

4MA1-1H Q24 Q24

MODULAR UNIT 1

4 marks

24 Express each of a , b and c in terms of q so that

$$q + 12x - qx^2$$

can be written as $a - b(x - c)^2$

 $a = \dots\dots\dots$ $b = \dots\dots\dots$ $c = \dots\dots\dots$

Question 16

4MA1-1H Q20 Q20(a), Q20(b)

MODULAR UNIT 1

4 marks

20 (a) Express $7 + 12x - 3x^2$ in the form $a + b(x + c)^2$ where a , b and c are integers.

.....
(3)

C is the curve with equation $y = 7 + 12x - 3x^2$

The point **A** is the maximum point on **C**

(b) Use your answer to part (a) to write down the coordinates of **A**

(.....,)
(1)

21 Express $5x^2 - 20x + 23$ in the form $a(x - b)^2 + c$ where a , b and c are integers.

(Total for Question 21 is 3 marks)

19 $ABCED$ is a five-sided shape.

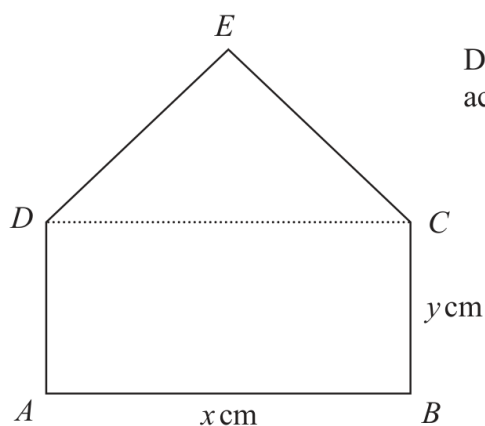


Diagram **NOT**
accurately drawn

$ABCD$ is a rectangle.

CED is an equilateral triangle.

$$AB = x \text{ cm} \quad BC = y \text{ cm}$$

The perimeter of $ABCED$ is 100 cm.

The area of $ABCED$ is $R \text{ cm}^2$

(a) Show that $R = \frac{x}{4} \left(200 - [6 - \sqrt{3}]x \right)$

(3)

(b) (i) Find the value of x for which R has its maximum value.

Give your answer in the form $\frac{p}{q - \sqrt{3}}$ where p and q are integers.

$$x = \dots\dots\dots$$

(2)

(ii) Explain why the maximum value of R is given by this value of x .

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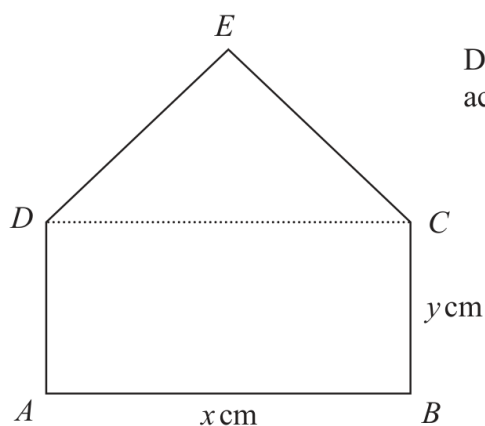


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